5. A stream of electrons travelling at a speed of 6.8×10^6 m s^{-1} and making an angle of 30° to the horizontal enters the region between two horizontal parallel plates at point P. The electrons are deflected and exit horizontally at point Q as shown in **Fig. 5.1**.

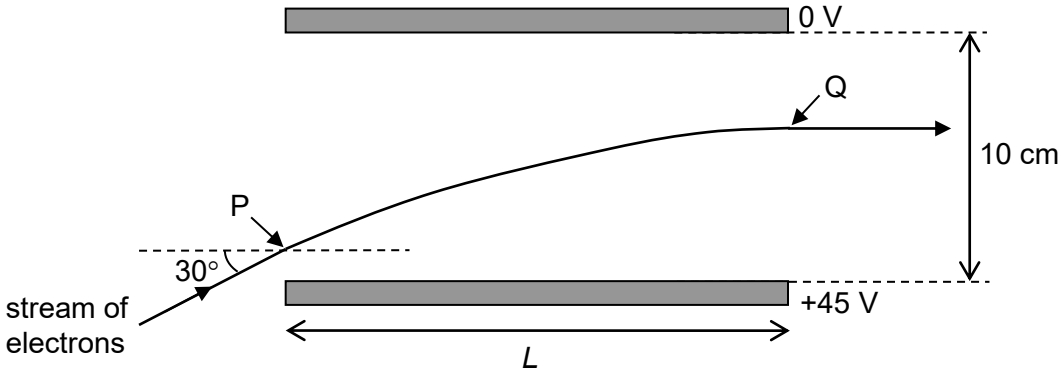

Fig. 5.1

The length of the plates is *L* and their separation is 10 cm. The potential of the lower plate is +45 V with respect to the upper plate.

(a) Show that the magnitude of the acceleration of an electron between the plates is 7.9×10^{13} m s^{-2}. [2]

(b) Determine the length L of the plates. [3]

(c) Charged particles travelling horizontally at a speed of 6.8×10^6 m s^{-1} are now entering the region between the plates at point Q as shown in **Fig. 5.2**.

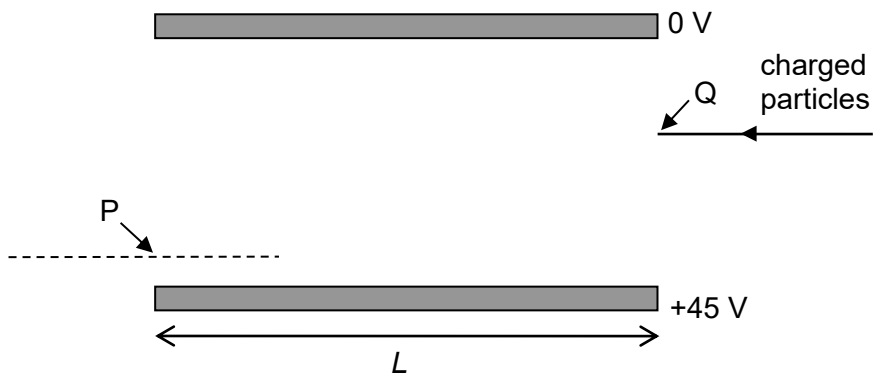

Fig. 5.2

(i) On **Fig 5.2**, sketch the path of the charged particles between the plates if they were electrons. Label this path A. [1]

(ii) On **Fig 5.2**, sketch the path of the charged particles between the plates if they were protons. Label this path B. [1]

(iii) Explain two differences between paths A and B. [2]

I.

II.